

Gender-Based Analysis of Cyclone Shelter Accessibility

1. Introduction

This report presents a technical analysis of the socio-economic and gender-specific factors influencing cyclone shelter accessibility in coastal Bangladesh. The study focuses on identifying why specific populations, particularly women, remain at risk despite the availability of physical infrastructure. The analysis utilizes data-driven visualizations to correlate demographic trends with behavioral outcomes.

2. Methodology

The research employed a mixed-methods approach, utilizing household surveys across multiple Unions. Data analysis was conducted using Python and Jupyter Notebooks, focusing on descriptive statistics and correlation matrices to identify significant barriers to evacuation.

3. Demographic and Vulnerability Profiling

A comprehensive understanding of the respondent profile is essential for contextualizing accessibility challenges.

- Gender Balance:** The survey captured a significant female perspective, with women making up 56.2% of the respondents, ensuring that gender-specific barriers are represented in the data.

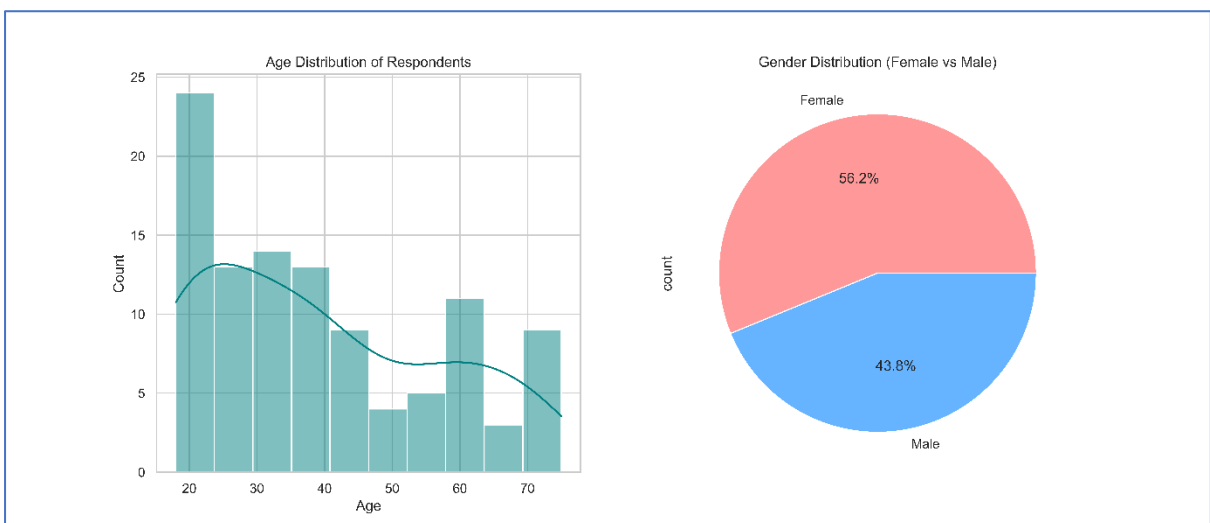


Figure 1: Age and Gender Distribution

- **Sanitation as a Barrier:** Analysis of age and gender distribution among those reporting "Poor Sanitation" shows that this concern spans all age groups, though it disproportionately impacts women's willingness to relocate during disasters.

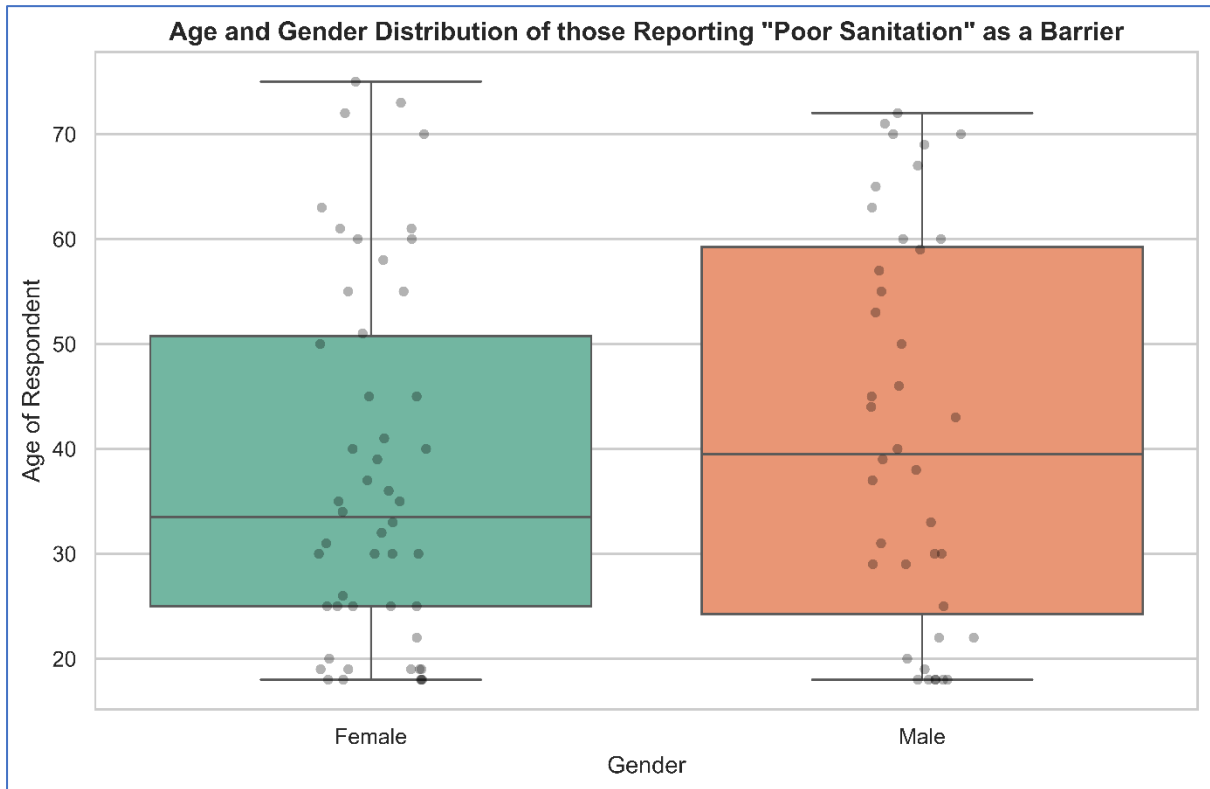


Figure 2: Is age and gender responsible for poor sanitation

4. Identification of Barriers to Shelter Access

The study categorized various deterrents that prevent residents from seeking safety.

- **Primary Deterrents:** "Overcrowding," "Lack of privacy or separate space," and "Poor sanitation" were identified as the most frequent mentions for avoiding shelters.

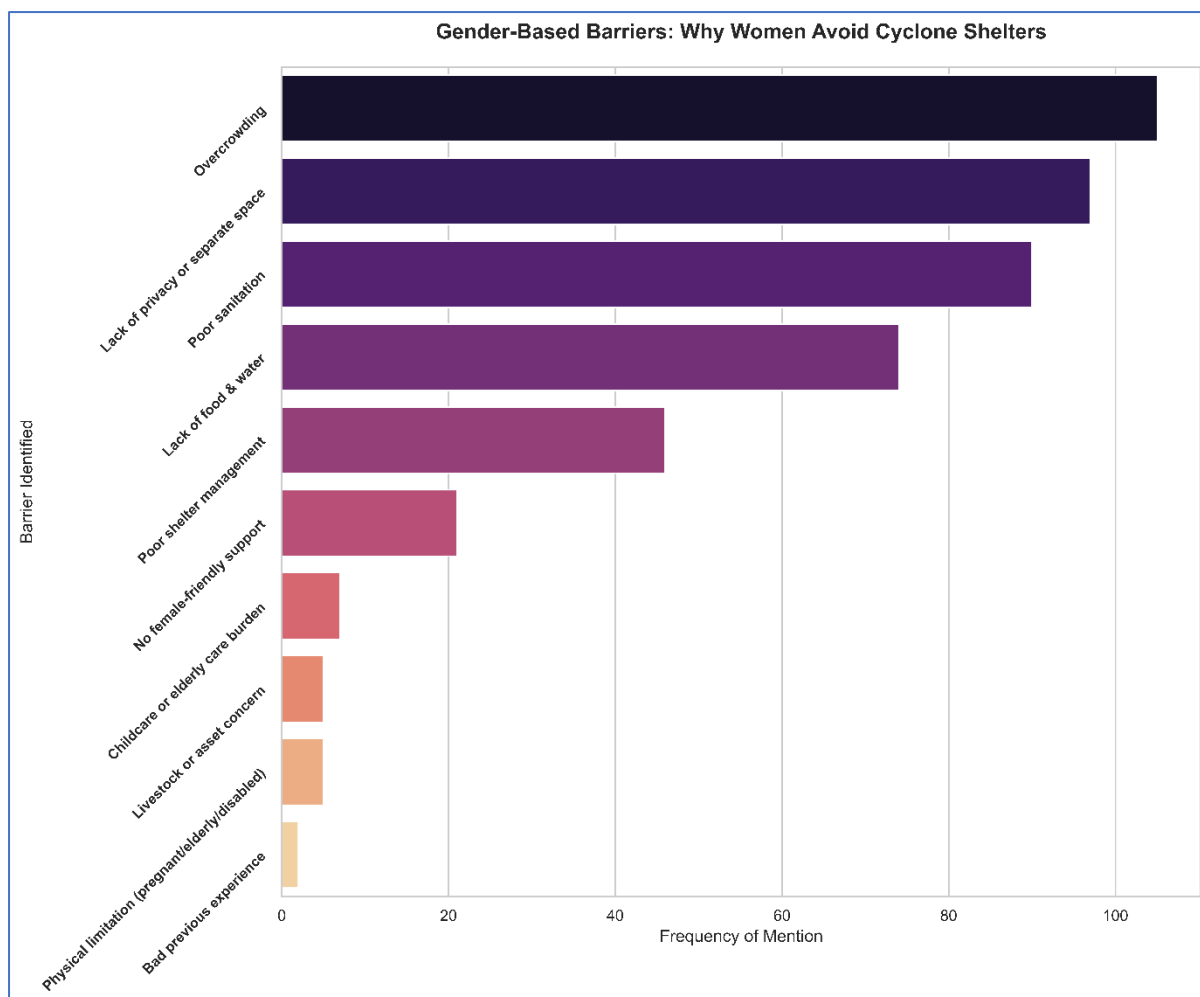


Figure 3: Gender based barriers

- Safety Perception:** Interestingly, while a majority of respondents feel that shelters provide physical safety, the qualitative environment—specifically privacy—creates a secondary layer of perceived risk for female occupants.

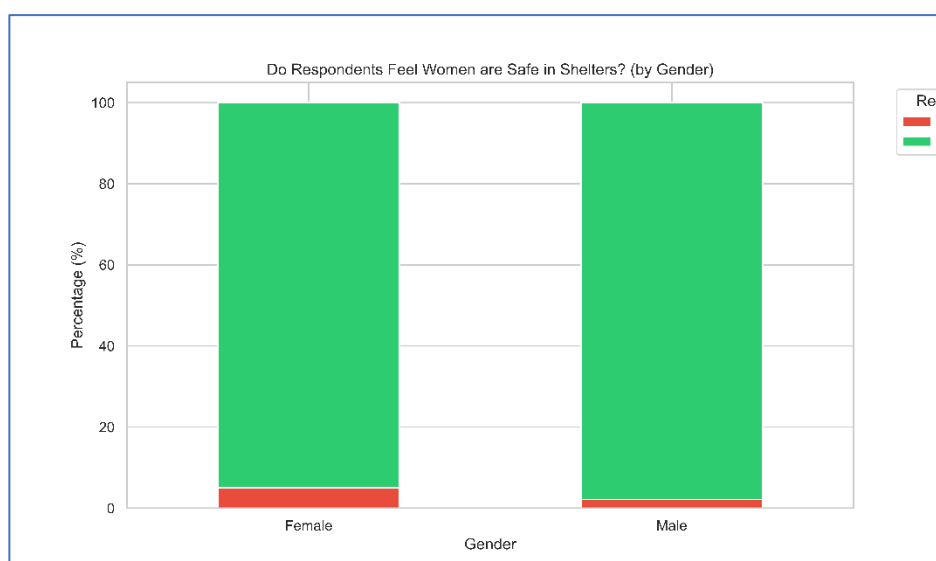


Figure 4: Is women feel safe in shelter?

5. Socio-Economic and Geographical Correlations

The ability to access a shelter is strongly linked to a household's economic status and geographic location.

- Economic Impact:** Respondents with a monthly household income of less than 5,000 BDT reported significantly higher difficulty ranks (Very Difficult) compared to higher-income brackets.

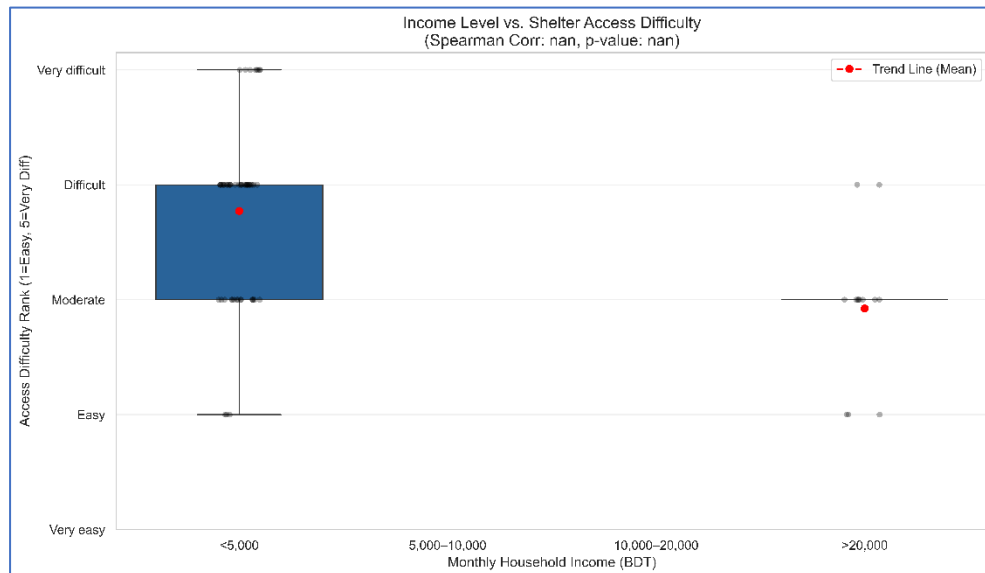


Figure 5: Relation between income and shelter access

- Union-Level Variation:** Accessibility is not uniform across the region. Unions such as **Sutobagi** and **Pachakoralia** exhibit higher levels of reported access difficulty compared to **Nishanbaria**.

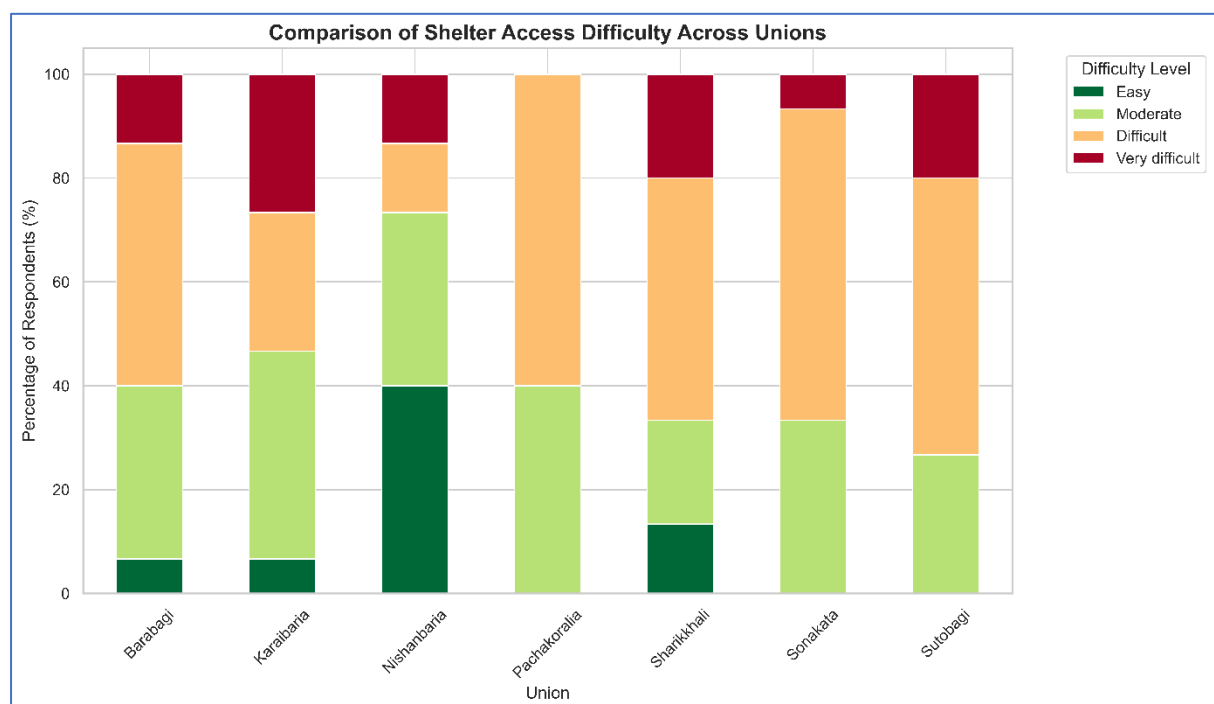


Figure 6: Comparison of shelter access difficulties

- Statistical Correlations:** A correlation matrix indicates a positive relationship between household size and access difficulty (0.46), suggesting that larger families face logistical bottlenecks during the evacuation window.

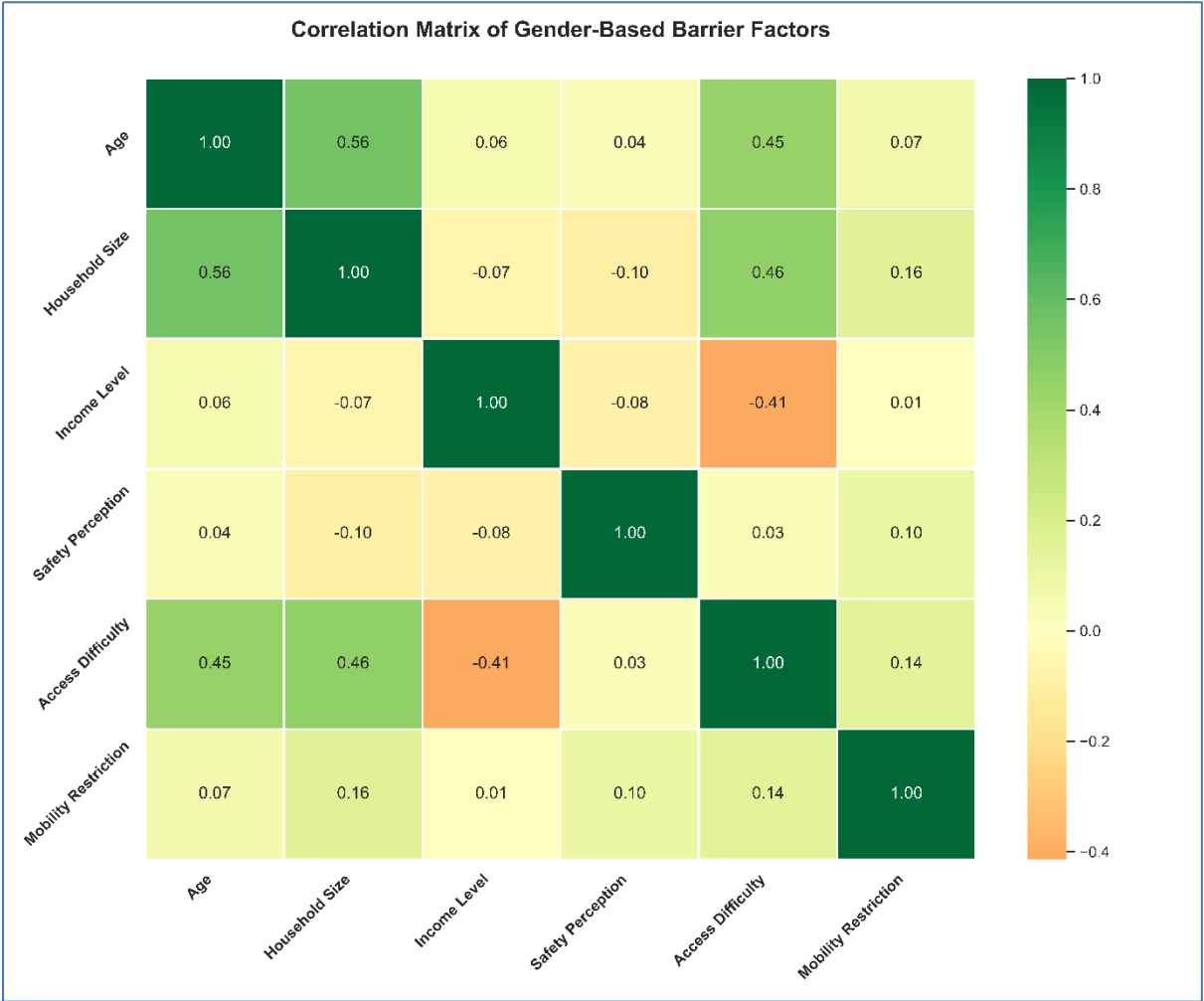


Figure 7: Correlations heatmap

6. Agency and Evacuation Decision-Making

The study analyzed the internal household dynamics that dictate whether a family evacuates.

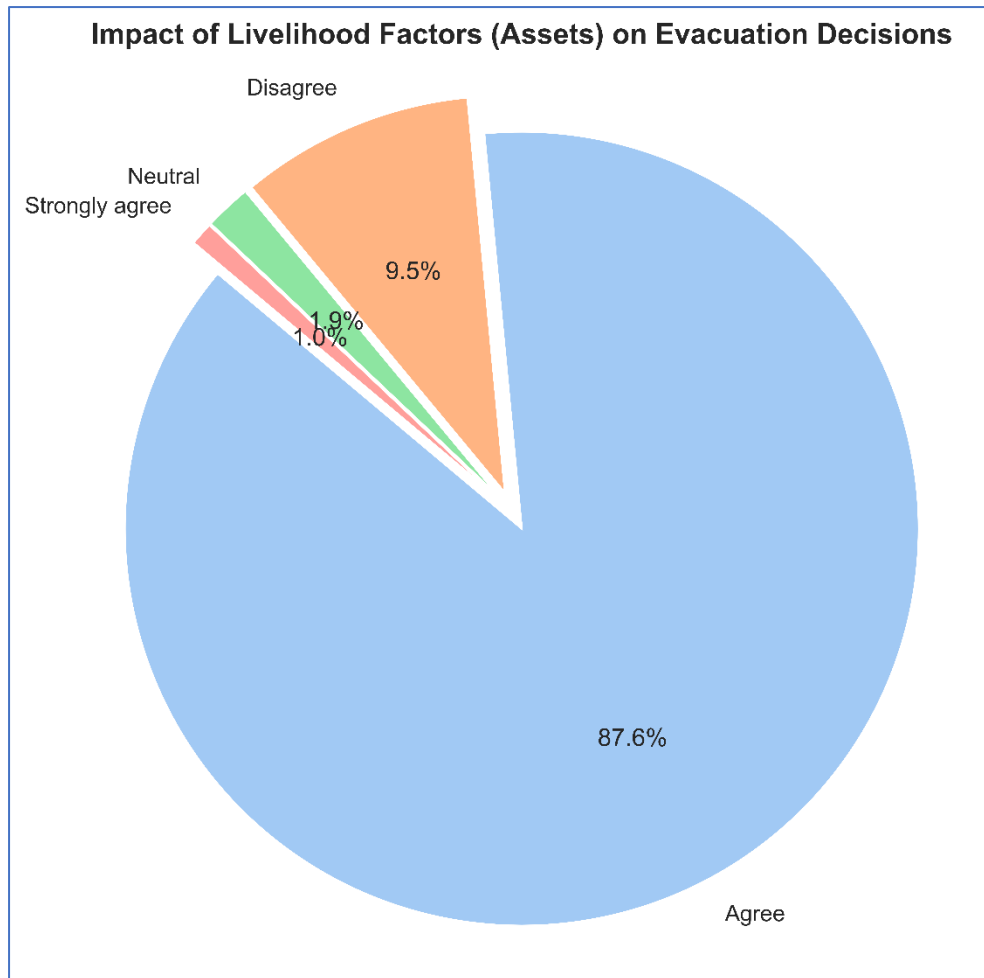


Figure 9: Livelihood factors that influence evacuation decisions

- **Livelihood Constraints:** 87.6% of respondents agreed that the need to protect assets and livestock is a primary reason for delaying evacuation, highlighting a critical conflict between survival and livelihood preservation.

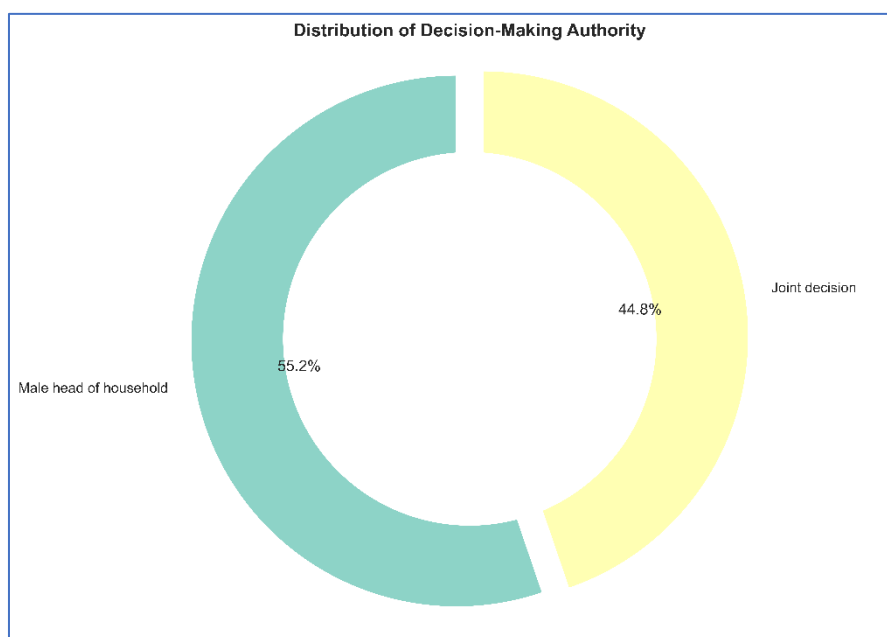


Figure 8: Distribution of decision making authority

- **Authority Distribution:** In the majority of cases (55.2%), the male head of household retains final decision-making authority regarding evacuation, although joint decision-making is present in approximately 44.8% of households.

7. Conclusion and Policy Recommendations

To increase the efficiency of disaster response and ensure gender-inclusive safety, the following measures are recommended:

1. **Infrastructure Gender-Mainstreaming:** Retrofit existing shelters with dedicated, private sanitation facilities and separate quarters for women and children.
2. **Asset Protection Strategies:** Integrate livestock shelters (Killas) within or near cyclone shelters to minimize the delay caused by asset concerns.
3. **Targeted Economic Support:** Prioritize accessibility improvements (e.g., road leveling or transport assistance) in Unions with the highest reported difficulty ranks.